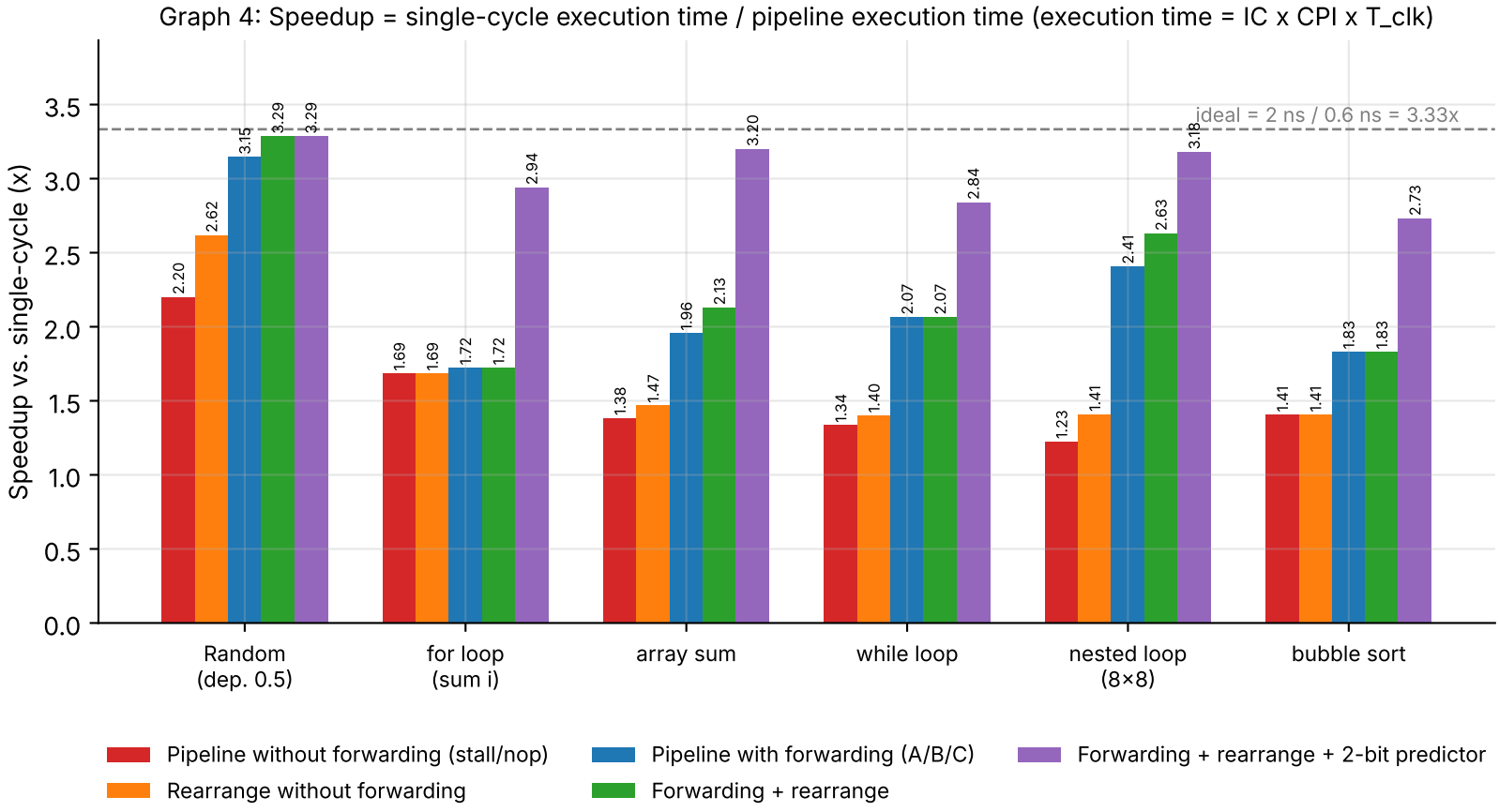
Graph 4: Speedup = single-cycle execution time / pipeline execution time (execution time = IC x CPI x T_clk)
Speedup vs. single-cycle (x)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
ideal = 2 ns / 0.6 ns = 3.33x
2.20
2.62
3.15
3.29
3.29
1.69
1.69
1.72
1.72
2.94
1.38
1.47
1.96
2.13
3.20
1.34
1.40
2.07
2.07
2.84
1.23
1.41
2.41
2.63
3.18
1.41
1.41
1.83
1.83
2.73
Random (dep. 0.5)
for loop (sum i)
array sum
while loop
nested loop (8×8)
bubble sort
Pipeline without forwarding (stall/nop)
Rearrange without forwarding
Pipeline with forwarding (A/B/C)
Forwarding + rearrange
Forwarding + rearrange + 2-bit predictor